Project Design Phase-II / Requirement Analysis

Solution Requirements (Functional & Non-Functional)

Date	28 October 2026
Team ID	Ujjawal Rathod (Individual Capstone)
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	4 Marks

Functional Requirements

Following are the functional requirements of the proposed Electric Vehicle Market Performance & Insights solution:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority
1	**Data Access & Integration**	Connect to and union `ev_market_train.csv` (200,000 rows) and `ev_market_test.csv` (100,000 rows) in Tableau Public to create a unified 300,000-row dataset across 37 attributes (including Vehicle ID, Manufacturer, Model Year, Category, Battery Type, Range Km, Efficiency Wh/Km, Customer Rating, Market Region, and Manufacturing Country).	**High**
2	**Data Processing**	Configure data types and roles: convert `Model Year` to Discrete dimension (2018–2026), assign Geographic Role (Country/Region) to `Manufacturing Country`, aggregate Range Km, Efficiency Wh/Km, and Customer Rating using Average, and allow Tableau to naturally exclude nulls (~1% in `annual_sales_units`, ~2% in `customer_rating`) without distorting calculations.	**High**
3	**Business Analysis**	Compute core analytical KPIs and custom fields: Total Sales = `SUM([Annual Sales Units])`, Sales (Millions), Percent of Total for Battery Tech Share, Distinct Count of Vehicle ID for production volume, and a FIXED Level of Detail expression `SUM([Annual Sales Units]) / MIN({FIXED : SUM([Annual Sales Units])})` to measure true Top 10 Manufacturer Market Share across all 200 manufacturers.	**High**
4	**Business Insights & Analytical Outputs**	Deliver 9 purpose-built worksheets (V1–V9), a unified 3×3 EV Market Dashboard (1400×1500 px), 3 persona mini-dashboards (Emily - Strategy, David - Competitive, Dr. Sarah - Research), and a 4-point interactive Tableau Story with data-backed insight captions.	**High**
5	**User Interaction & Analysis**	Provide global multi-select dropdown/slider filters for Market Region, Battery Type, and Model Year applied to all worksheets using the data source, alongside interactive click-to-filter actions on V1 (Category Bars), V4 (Battery Donut), and V7 (Manufacturing Country Map).	**High**
6	**Data Update & Reporting**	Publish the packaged analytical workbook (300k rows embedded) to Tableau Public and integrate the live Dashboard and Story iframes into a Python Flask web	**Medium**

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority
		application (`app.py` + `index.html`) deployed publicly on Vercel and documented on GitHub.	

Non-Functional Requirements

Following are the non-functional requirements of the proposed solution:

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Intuitive 3×3 tiled dashboard layout (1400×1500 px) and 4-point Story navigator with clear titles, millions (M) number formatting, non-overlapping legends, and zero-anchored axes (0–400 km on V6, 0–5 on V8) to prevent visual misinterpretation.
NFR-2	**Security**	Read-only public visualization access hosted over HTTPS via Tableau Public and Vercel cloud infrastructure, ensuring underlying raw CSV records cannot be altered or overwritten by end users.
NFR-3	**Reliability**	Accurate aggregation across 300,000 rows where missing values (~1% in sales, ~2% in ratings) are safely ignored by Tableau's SUM/AVG engine, and Top-10 market share remains mathematically accurate via FIXED LOD expressions.
NFR-4	**Performance**	Initial dashboard and story load time under 3 seconds for 300,000 records, with interactive global filters and click-to-filter actions updating all 9 visuals within 1 to 2 seconds.
NFR-5	**Availability**	24/7 global browser availability through Tableau Public cloud servers and serverless Python Flask hosting on Vercel, accessible across desktop and mobile browsers without local software installation.
NFR-6	**Scalability**	Wildcard/manual union architecture and modular Flask template design allow seamless ingestion of additional model years, regions, or EV datasets beyond 300,000 rows without redesigning the schema.